Logarithmic inequalities

Domain, then direction, then solve — and the intersection at the end that catches everything.

LESSON 65–66 | 2 × 40 MIN | ALGEBRA 10, §3.8 | P2 · 2.4

LESSON CLOCK

12' | 22' | 22' | 22' | 20' | 12'

Three steps, in order	12 min
Base above 1	22 min
Base below 1	22 min
Substitution	22 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- State the domain of a logarithmic inequality before solving.
- Use monotonicity to compare arguments, reversing when the base is below 1.
- Intersect the solution with the domain.
- Solve inequalities that become quadratic after a substitution.

TEXTBOOK

National	pp. 273–284
Cambridge	Pure Mathematics 2 & 3, pp. 39–44
Workbook	P2 Exercise 2G

01 Explanation

Three steps, in order

NEVER CHANGE THE ORDER

1 Write the domain — every argument strictly positive. **2** Compare the arguments, reversing the sign if the base is under 1. **3** **Intersect** the result with the domain.

Step 3 is not optional and is not a check: it is where a large part of the answer is decided. An inequality solved correctly and not intersected is usually wrong.

Base above 1

$$a > 1:\ \log_a f > \log_a g \Leftrightarrow f > g > 0$$

Example. $\log_2(x-1) < 3$.

1. Domain: $x > 1$.
2. $\log_2(x-1) < \log_2 8$, base > 1, so $x - 1 < 8$ and $x < 9$.

3. Intersect: $1 < x < 9$.

Without step 3 the answer would be $x < 9$, which includes $x = 0$ — where the expression does not exist.

Base below 1

$$0 < a < 1: \log_a f > \log_a g \Leftrightarrow 0 < f < g$$

Example. $\log_{1/3}(x + 2) > -1$.

1. Domain: $x > -2$.
2. $-1 = \log_{1/3} 3$. Base < 1, so reverse: $x + 2 < 3$ and $x < 1$.
3. Intersect: $-2 < x < 1$.

TWO THINGS REVERSE, NOT ONE

A base under 1 reverses the comparison of the arguments — and if the inequality is later multiplied by a negative or divided, that reverses again. Do one reversal at a time, and say out loud which one you are doing.

Substitution

With $t = \log_a x$ (no sign condition) the inequality becomes an ordinary quadratic one:

$$(\lg x)^2 - 3 \lg x + 2 < 0 , t = \lg x \Rightarrow 1 < t < 2$$

$$1 < \lg x < 2 \Rightarrow 10 < x < 100$$

The base 10 exceeds 1, so the order is preserved. With a base under 1 the whole double inequality would flip.

THE THREE SOURCES OF ERROR, IN ORDER OF FREQUENCY

forgetting the intersection · forgetting the reversal · forgetting that t has no sign condition. Address them in that order.

02 Worked examples

EXAMPLE 1 **Solve $\log_2(x - 1) < 3$.**

1. Domain $x > 1$.
2. $x - 1 < 8$

   Base > 1.
3. $x < 9$
4. Intersect.

ANSWER

$1 < x < 9$

EXAMPLE 2 **Solve $log\_(1/3)(x + 2) > -1$.**

1. Domain $x > -2$.
2. $-1 = log\_(1/3)\ 3$
3. Base < 1: reverse. $x + 2 < 3$.
4. $x < 1$; intersect.

ANSWER

$-2 < x < 1$

EXAMPLE 3 **Solve $\lg x + \lg(x-3) < 1$.**

1. Domain $x > 3$.
   Both arguments.
2. $\lg[x(x-3)] < \lg 10$
3. $x^2 - 3x - 10 < 0 \Rightarrow -2 < x < 5$
4. Intersect with $x > 3$.

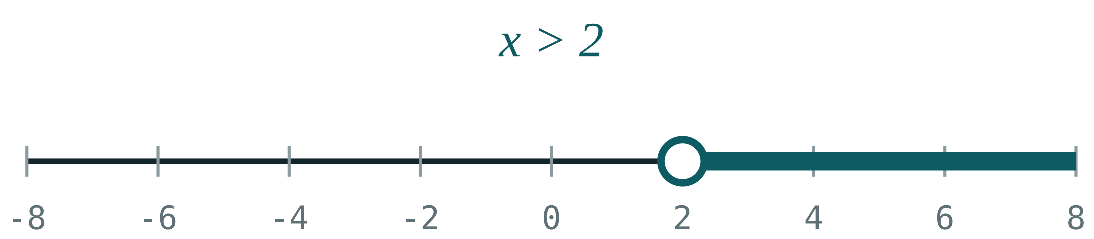
ANSWER

$3 < x < 5$

03 Interactive model

Draw two number lines — the domain and the solution — and shade the overlap.

Domain and solution



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Logarithmic inequality	Logarifmik tengsizlik	Логарифмическое неравенство
Domain of definition	Aniqlanish sohasi	Область допустимых значений
Monotonicity	Monotonlik	Монотонность
Reversing the sign	Ishorani almashtirish	Смена знака
Intersection with the domain	Soha bilan kesishma	Пересечение с ОДЗ
Substitution	Almashtirish	Замена
Solution set	Yechimlar to‘plami	Множество решений
Interval notation	Oraliq belgilanishi	Интервальная запись

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first step is:

- compare arguments
- write the domain
- substitute
- reverse

$log_2(x - 1) < 3$ gives:

- $x < 9$
- $1 < x < 9$
- $x > 1$
- $x < 8$

A base under 1:

- keeps the sign
- reverses the sign
- makes it undefined
- doubles it

In $t = lg\ x$ the condition on t is:

- $t > 0$
- none
- $t \geq 0$
- $t \neq 0$

$lg\ x < 2$ gives:

- $x < 100$
- $0 < x < 100$
- $x > 100$
- $x < 2$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $lg\ x > 1$

   ANSWER $x > 10$

2. Solve $lg\ x < 2$

   ANSWER $0 < x < 100$

3. Solve $log_2\ x > 3$

   ANSWER $x > 8$

4. Solve $log_2\ x < 0$

   ANSWER $0 < x < 1$

5. Domain of $log_3(x - 5)$

   ANSWER $x > 5$

6. Solve $log_5\ x \geq 1$

   ANSWER $x \geq 5$

7. Solve $log\_(1/2)\ x > 0$

   ANSWER $0 < x < 1$

Medium · the standard question

7 PROBLEMS

1. Solve $log_2(x - 1) < 3$

   ANSWER $1 < x < 9$

2. Solve $log_3(2x + 1) > 2$

   ANSWER $x > 4$

3. Solve $log\_(1/3)(x + 2) > -1$

   ANSWER $-2 < x < 1$

4. Solve $lg\ x + lg(x - 3) < 1$

   ANSWER $3 < x < 5$

5. Solve $log_2(x + 3) \leq 4$

   ANSWER $-3 < x \leq 13$

6. Solve $log\_(0.5)(x - 1) \geq -2$

   ANSWER $1 < x \leq 5$

7. Solve $log_4(x^2) < 1$

   ANSWER $-2 < x < 2, x \neq 0$

Hard · stretch and reason

7 PROBLEMS

1. Solve $(\lg x)^2 - 3 \lg x + 2 < 0$

ANSWER $10 < x < 100$

2. Solve $(\log_2 x)^2 - \log_2 x - 6 \geq 0$

ANSWER $0 < x \leq \frac{1}{4}$ or $x \geq 8$

3. Solve $\log_2(x - 1) + \log_2(x + 1) < 3$

ANSWER $1 < x < 3$

4. Solve $log\_(1/2)(x^2 - 3x) \geq -2$

ANSWER $-1 \leq x < 0$ or $3 < x \leq 4$

5. Solve $\log_3(x + 1) > \log_3(2x - 3)$

ANSWER $1.5 < x < 4$

6. Solve $\lg(x - 2) < \lg(4 - x)$

ANSWER $2 < x < 3$

7. Explain why $\log_2 x < 3$ is not simply $x < 8$

ANSWER The domain $x > 0$ must be intersected

07 Homework

Homework — 6 tasks

Draw the domain and the solution on the same number line every time.

1. Solve $\log_3(x - 2) < 2$.
2. Solve $log\_(1/4)(x + 1) > -1$.
3. Solve $\lg x + \lg(x - 21) < 2$.
4. Solve $(\log_3 x)^2 - 4 \log_3 x + 3 \leq 0$.
5. Solve $\log_2(3x - 1) \geq \log_2(x + 5)$.

6. Explain in three sentences why the intersection with the domain is part of the method and not a check.